

FMN050 Numerical Analysis for E3

Solutions for Homework assignment 3

1. Linear Algebra

To see when the system has unique solutions is enough look when the matrix A is non-singular. This is $0 \neq \det(A) = -3z(y+3) \Leftrightarrow z \neq 0$ and $y \neq -3$. In other the hand, the system can have infinitely many solutions or no solution when the opposite happens. From here, there are two cases to analyze:

Case I: $z = 0$.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 3 & -5 & 0 \\ 1 & 7 & 2 & 0 \\ y & 10 & y & 0 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 2 \\ -2 \\ 6 \\ y \end{pmatrix}$$

Hereinafter, making some linear algebra manipulations, the system looks like

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & -5 & 0 \\ 0 & 7 & 2 & 0 \\ 0 & 10 & y & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \\ -y \end{pmatrix}$$

Hence, the values of x_2 and x_3 are determined by the system

$$\begin{pmatrix} 3 & -5 \\ 7 & 2 \end{pmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

Therefore $x_2 = \frac{24}{41}$ and $x_3 = -\frac{2}{41}$. Then $y = -\frac{80}{13}$, otherwise the system has no solution.

Case II: $y = -3$. Setting the value of y the system becomes in

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 3 & -5 & 0 \\ 1 & 7 & 2 & z \\ -3 & 10 & -3 & z \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 6 \\ -3 \end{pmatrix}.$$

After some algebraic manipulation the system becomes in

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & -5 & 0 \\ 0 & 7 & 2 & z \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \\ -3 \end{pmatrix}$$

that is clearly inconsistent.

Finally the system $Ax = b$:

- a. Has unique solution if $z \neq 0$ and $y \neq -3$.
- b. Has infinitely many solutions if $z = 0$ and $y = -\frac{80}{13}$.
- c. Has no solution if $z = 0$ and $y \neq -\frac{80}{13}$, or $z \in \mathbb{R}$ and $y = -3$.

2. Condition of the Vandermonde system

Task 2a): MATLAB-function `vandercond.m`:

```
close all
clear all

condnumbers = zeros(1, 35);
for n = 1:35
    t = linspace(-1, 1, n);
    condnumbers(n) = cond(vander(t));
end

semilogy(1:35, condnumbers)
grid
title('Condition numbers of Vandermonde matrices')
xlabel('Matrix dimension (square)')
ylabel('log(kappa(V))')
```

According to theory the dependence is at least exponential. The figure indicates exponential dependence, with $\kappa(V) \approx e^{n-1}$ (look at the start values also).

Task 2b):

The relative error in output is $\kappa(V)|_{n=16} \cdot \epsilon_{mach} \approx e^{16-1} \cdot 2^{-53} = 4 \cdot 10^{-10}$.

3. An ill-conditioned linear system

Task 3a):

If $A = LDL^T$ and $Ax = b$, then

$$L \underbrace{DL^T}_y x = b. \quad (1)$$

This is solved as follows:

1. Let $y = DL^T x$ and compute y by solving

$$Ly = b. \quad (2)$$

As this system is lower triangular, this is done by *forward substitution*.

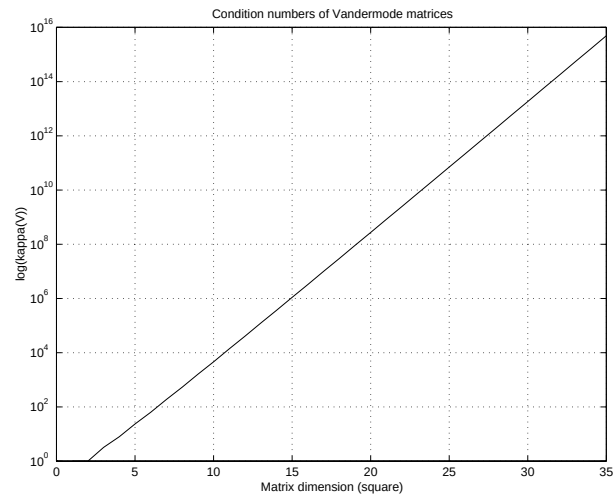


Figure 1: Condition numbers of the Vandermonde matrices with dimension $1 \dots 35$.

2. Compute $z = L^T x$ by solving $Dz = y$. As D is diagonal, this is just one division for each component.
3. Compute x by solving

$$L^T x = z. \quad (3)$$

As L^T is upper triangular, use *backward substitution*.

Task 3b): If R is a right upper triangular matrix, $R^T R x = b$ can be solved as follows

1. Let $y = R x$ and solve $R^T y = b$ by forward substitution.
2. Compute x by solving $R x = y$ via backward substitution.

Task 3c):

The MATLAB file `hilbertsys.m`:

```
close all
clear

% Dimension.
n = 12;

% The Hilbert matrix of this dimension.
H = hilb(n);
```

```

% The condition number of the Hilbert matrix
% in inf-norm.
Hcond = cond(H, inf)

% The right hand side is the same as the
% rightmost column of H.
b = H(:, n);

% Cholesky factorization.
R = chol(H);

% Solution of the equation.
y = R' \ b;
x = R \ y;

% The analytical solution.
I = eye(n);
x_analytic = I(:, n);

% Norms and errors of this solution.
norm_error = norm(x - x_analytic, inf)
norm_x_analytic = norm(x_analytic, inf)
rel_error = norm_error / norm_x_analytic

expected_error = Hcond * 2^(-53)

```

The relative error is estimated to 0.0076. NB: This error was calculated by a PC running SuSe Linux. On other computer systems, the results may vary slightly. We expect from theory that

$$\frac{\|\hat{x} - x\|_{\infty}}{\|x\|_{\infty}} \approx \kappa_{\infty}(H) \cdot \epsilon_{mach} = 4.0492 \cdot 10^{16} \cdot 1.1102 \cdot 10^{-16} = 4.4955. \quad (4)$$

4. Average property of Least Squares

We want to adjust a straight line $y = mx + b$ after N measure points $(x_i; y_i)$ with $i = 0, \dots, N$. We formulate the problem as the following linear system

$$\underbrace{\begin{pmatrix} y_n \\ \vdots \\ y_1 \end{pmatrix}}_{y:=} = \overbrace{\begin{pmatrix} x_n & 1 \\ \vdots & \vdots \\ x_1 & 1 \end{pmatrix}}^{A:=} \underbrace{\begin{pmatrix} m \\ b \end{pmatrix}}_{x:=}.$$

We have identified all the parts of this overdetermined linear system. Now we use the normal equations to solve it in the least squares sense obtaining

$$A^T y = A^T A x \Rightarrow \begin{pmatrix} \sum_{i=1}^n x_i y_i \\ \sum_{i=1}^n y_i \end{pmatrix} = \begin{pmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix}.$$

The matrix $A^T A$ can be inverted by hand to finally obtain the slope m and the intersect b . It results in

$$\begin{pmatrix} m \\ b \end{pmatrix} = \frac{1}{n \sum x_i^2 - (\sum x_i)^2} \begin{pmatrix} n \sum x_i y_i - \sum \sum y_i x_k \\ - \sum \sum x_k x_i y_i + \sum \sum x_i^2 y_k \end{pmatrix}.$$

Now with this values, we will verify that the point (x^*, y^*) lies in the line $y = mx + b$. Substituting x^*, m and b and then with some algebraic manipulations we get that

$$\begin{aligned} mx^* + b &= \frac{1}{n \sum x_i^2 - (\sum x_i)^2} \left(\sum_{i,k} x_i y_i x_k - \frac{1}{n} \sum_{i,j,k} y_i x_k x_j - \sum_{i,k} x_k x_i y_i + \sum_{i,k} x_i^2 y_k \right) \\ &= \left(\frac{1}{n} \sum y_i \right) \frac{1}{n \sum x_i^2 - (\sum x_i)^2} \left(n \sum x_i^2 - \sum_{k,j} x_k x_j \right) \\ &= y^*. \end{aligned}$$

This means that the point (x^*, y^*) indeed lies in the line.

5. Widlar source current

We observe that the equation governing the Widlar source current is linear in the temperature. We can therefore apply least squares to find a possible temperature. We construct the classical linear system $AV_t = b$ with $b = (Re_i * I_{o_i})_i$ and $A = (\ln \frac{I_{ref_i}}{I_{o_i}})_i$. Notice that the matrix A is a column vector since we only have one unknown variable. Now, the following sequence of commands in *Matlab*'s prompt solves the problem.

```
>> Io=[7.859763 11.210700 11.165746 7.472788 9.294492 13.264235]'*1e-6;
>> Iref=[1 2 3 10 12 17]'*1e-3;
>> Re=[16 12 13 25 20 14]'*1e3;
>> b=Io.*Re;
>> A=log(Iref./Io);
>> format long
>> Vt=A'*A\A'*b
```

```

Vt =

    0.02595052479272

>> K=1.38066e-23;
>> q=1.60218e-19;
>> T=Vt*q/K

T =

    3.011415686150070e+002

>> T-273-25

ans =

    3.14156861500703

>> pi

ans =

    3.14159265358979

>> norm(A*Vt-b)

ans =

    6.971403378297082e-009

```

Notice that the residual is around 5 orders of magnitude below the answer, which tells us that the solution is good, maybe for our purposes. The comparison with the number π is because the temperature in the room where the tests were done was exactly $25 + \pi$ Celsius (another instrument was used to assure that!). We observe that after all we recover 5 significant digits of π , that makes sense if we look at the order of the residual.